VIMALAN ARUNPRAGASH

SOFTWARE ENGINEER INTERN

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PROFILE SUMMARY

IT undergraduate passionate about creating innovative solutions and improving intelligent systems. Excel at solving problems by breaking down complex challenges and delivering effective, practical solutions. Enjoy working collaboratively in teams, leveraging strong communication skills and a flexible mindset to contribute to project success.

SKILLS

Programming Languages : Python, Java, C, JavaScript, SQL

Machine Learning : TensorFlow, PyTorch, Keras, Scikit-learn, Transformers, ANN

NLP & Computer Vision : HuggingFace Transformers, BERT, GPT, NLTK, SpaCy, OpenCV, YOLO

Data Analysis & Visualization : Pandas, NumPy, Matplotlib, Seaborn, Plotly

Cloud & Development Tools : AWS, Google Cloud, Docker, Git, CI/CD, PostgreSQL, MongoDB

Web Development : Django, FastAPI, REST APIs

Other : Data Structures, Algorithms, Jupyter Notebooks, Streamlit

PROJECTS

AI-Driven Real-Time Learning & Assessment Platform

An AI-powered interview simulation platform leveraging speech-to-speech technology and multiple AI personas to conduct, evaluate, and provide real-time feedback.

- Designed and developed a real-time online learning platform with instructor–student interaction.
- Implemented AI-based student engagement and performance prediction using machine learning models.
- Built real-time features such as live sessions, automatic question triggering, and activity tracking using WebSockets.
- Developed instructor dashboards for session management, material uploads, and student monitoring.
- Integrated secure authentication, role-based access control, and scalable backend APIs.
- Improved learning effectiveness by providing automated feedback and analytics for instructors.

Technologies Used: React, TypeScript, FastAPI, Python, Machine Learning, WebSockets

Student Performance & Grade Prediction System

A machine learning system to predict final student grades using academic and behavioral features.

- Applied data preprocessing, outlier handling, feature engineering, and label encoding techniques.
- Tuned neural network architectures using Bayesian Optimization (Optuna) to improve accuracy.
- Evaluated models using accuracy, loss, confusion matrix, and classification reports.
- Implemented ensemble learning with soft voting to enhance prediction robustness.

Technologies Used: Python, ANN, Optuna, Scikit-learn, TensorFlow

ChefBuddy – Android Meal Planning Application

An Android mobile application to assist users with meal planning, recipe discovery, and food management.

- Implemented MVP architecture to ensure clean separation of concerns and maintainable code.
- Integrated TheMealDB API using Retrofit and Gson to fetch recipes by ingredients, category, and region.
- Designed features for weekly meal planning, allowing users to schedule meals using a calendar interface.
- Enabled users to save favorite recipes locally using Room (SQLite) for offline access.
- Displayed detailed step-by-step cooking instructions with ingredient lists, images, and tips.
- Built a modern, responsive UI using Material Design, Navigation Component, Glide, and Lottie animations.

Technologies Used: Java, Android, MVP Architecture, TheMealDB API

ACHIEVEMENTS

- RUSL Mini Hackathons participant
- IEEE Xtreme 17 – Sri Lankan Rank 72 (Global: 885)

- Coderally participant
- Winners – IEEE Innovation Nation Sri Lanka (INSL) 2025; represented the North Central province
- IEEE Student Branch of Rajarata University of Sri Lanka – Web Design Team Member (2024)

EDUCATION

Bachelor of Science in Information Technology (GPA: 2.41)
Rajarata University of Sri Lanka | 2022 – Present

G.C.E. Advanced Level Examination
Jaffna Hindu College | 2017 – 2020
Combined Maths – A, Physics – S, Chemistry – C (Z-Score: 0.7)

LANGUAGES

- English
- Tamil

OTHER SKILLS

- Teamwork
- Creativity
- Communication skills
- Problem solving